When a Control Does Not Control: A Mechanism-Derived Refutation of a False Positive in p-adic Cryptanalysis of the ECDLP

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Revised version (v3). Code and data: `ecdsa-lift` repository, commit `674750a`.

Abstract

Automated pipelines that fan out many structural probes against a hard problem and surface the most anomalous are false-positive engines: with enough encodings and transforms an apparent 5–11σ separation from a generic random baseline is cheap. This paper is a methodological case study of one such false positive, fully dissected. In a systematic negative exploration of p-adic lifting attacks on the elliptic curve discrete logarithm problem (ECDLP), a Gowers U^2 / Fourier analysis of the Teichmüller lift error $\delta(k)$ produced a persistent elevation (standardized effect sizes ∼3–14 on a clean controlled set, reaching ≈ 44 on a confirmatory curve with $n \approx 10^4$; throughout, these are effect sizes against a 300-draw empirical null, *not* Gaussian tail probabilities) that a standard value-permutation control certified as "real, index-ordered structure." We show this was a *control that cannot separate the hypotheses it was meant to test.* Mechanism-derived nulls—one per candidate explanation—together with a *positive* control localize the entire signal to a single exact, public identity, the antisymmetry $\delta(k)+\delta(n-k) = [n]\tau(G)$: synthetic sequences carrying *only* that reflection reproduce the elevation and its growth with n, and—a new converse test—a section deliberately chosen to break the reflection collapses the elevation to baseline. We make the mechanism analytic. The reflection turns each Fourier mode into an *improper* complex Gaussian, and the identity $\mathbb{E}|Z|^4 = 2\sigma^4 + |\rho|^2$ for pseudo-variance ρ yields a closed-form classification: a linear constraint inflates the U^2 mass by a factor $1 + \langle|\rho|^2\rangle/2\sigma^4 \in [1, 3/2]$, maximal $(3/2)$ exactly for index-reversal, giving a standardized growth $z(\text{syn/rand}) = 2^{7/4}(3^{1/4} - 2^{1/4})\sqrt{L} \approx 0.427\sqrt{L}$ that an auditor can apply directly. An inertness theorem—an explicit instance, in lift-error language, of a principle of Silverman and of Gadiyar–Padma, and which we state at the level of an arbitrary abelian-group extension—explains why every such anomaly on these curves must reduce to artifact-or-identity: for $\gcd(n, p) = 1$ the only secret-dependent part of a section lift error is $[k]c$, governed by $k \bmod p^{e-1}$, which the public data (E, G, Q) does not determine over the natural lift (and for which, as a separate state-of-knowledge remark, no efficient algorithm is known). The single inequality $\gcd(n, p) = 1$ is sharp: when it fails ($n = p$, anomalous curves) the *ambiguity* is annihilated and $[p]\tau(G)$ becomes a section-independent, secret-linear invariant recovering k with 100% success—Smart's attack. The experiments are diagnostic, not an empirical security evaluation of deployed curves; the reach to large-prime ECDSA-type curves is through the theorem's generality, not through cryptographic-scale experiments. The transferable contribution is the discipline: a control is valid only against a named alternative hypothesis, nulls must be derived from mechanisms, a positive control must reproduce any surviving signal, and a structural reason for inertness beats an accumulation of passed tests.

*Correspondence: william.todd.trevena@gmail.com. Code, data, and a per-phase index: the `ecdsa-lift` repository (release tagged with this paper).

1 Introduction

A recurring temptation in cryptanalysis is to lift a hard problem into a richer setting—a p-adic ring, a formal group, a higher-genus Jacobian—hoping that structure invisible over $\mathbb{F}_p$ becomes visible and exploitable. For the ECDLP this hope is well mapped: Silverman's "four faces of lifting" [2, 1] catalogues the local/global $\times$ torsion/non-torsion strategies and the obstruction each meets, and the only p-adic local lift that ever broke an ECDLP is Smart's attack on anomalous curves [4, 5, 6], where the group order equals the characteristic.

Automating such explorations sharpens an old hazard. A pipeline that computes many encodings of a lift invariant and reports the most unusual will, after enough nulls, return an "anomaly," and the analyst must decide whether it is signal. We conducted a long, deliberately exhaustive negative exploration of the canonical-lift attack surface for ordinary curves with large prime-order subgroups—the ECDSA regime—and hit exactly this case: a higher-order Fourier probe returned a clear 5–11σ effect on the original sweep, a textbook control endorsed it, and only a more careful experiment overturned it. The overturning, and the reason the anomaly *had* to be spurious, are the substance of this paper.

Contributions.

- A worked, quantitative example of a *control that does not control*: a permutation test that appears to certify structure but provably cannot separate the competing explanations (§4).
- *Mechanism-derived controls*: enumerate the candidate generating mechanisms and build one null per mechanism, including a *positive* control in which a synthetic model of the surviving mechanism reproduces the statistic quantitatively, and a *converse* control—a section deliberately chosen to break the antisymmetry, on which the elevation collapses (§5). This localizes the anomaly (an elevation of standardized size ∼3–14 on the clean controlled set) to one exact public identity.
- A closed-form account of the artifact (§5, Prop. 1): a pseudo-variance identity that derives the mass-inflation factor of *any* index-coupling constraint, recovering the reflection's exact 3/2 and the $\approx 0.427\sqrt{L}$ growth law as the maximal case, and predicting (via a carry model) the small residual spectral mismatch as a falsifiable effect at $L \gtrsim 3000$.
- An inertness theorem (§7), stated for an arbitrary abelian-group extension, explaining why such anomalies are forced: for $\gcd(n, p) = 1$ the secret-dependent part of any section lift error is $[k]c$, governed by $k \bmod p^{e-1}$, which the public data (E, G, Q) does not determine over the natural lift (and which no known efficient algorithm computes from it—a state-of-knowledge remark, kept separate from the theorem). We present it as the explicit instance it is of [2, 3], with proof and verification.
- A sharp-boundary demonstration (§8): the same construction on an anomalous curve ($\gcd(n, p) = p$) recovers the secret with 100% success (52/52 over all nonzero residues). The theorem's one hypothesis is exactly the line between inert and broken.
- A scope map (§11): precisely which attack schema this refutes, which it does not, and a protocol future searches should follow.

The theorem is not new mathematics; its role is to certify, after the fact, that the methodological suspicion was structurally warranted. We stress at the outset that the experiments are diagnostic:

they expose and explain a false positive in a lift-error statistic, they are not performed at cryptographic sizes, and they constitute no empirical security evaluation of any deployed ECDSA curve. The extension to ordinary prime-order ECDSA-type curves is carried by the structural theorem, which needs only $\gcd(n, p) = 1$, not by the curve sizes tested here.

2 Background and notation

Let $p \geq 5$ be prime and $E/\mathbb{F}_p$ an elliptic curve with $\#E(\mathbb{F}_p) = N$. Fix $G \in E(\mathbb{F}_p)$ of order n with $\gcd(n, p) = 1$, and $Q = kG$; the ECDLP is to recover $k \in \mathbb{Z}/n\mathbb{Z}$ from (E, G, Q). For $e \geq 1$, reduction modulo p on a fixed Weierstrass lift gives the exact sequence of abelian groups

$$0 \longrightarrow \widehat{E}(p\mathbb{Z}/p^e) \longrightarrow E(\mathbb{Z}/p^e) \xrightarrow{\text{red}} E(\mathbb{F}_p) \longrightarrow 0, \tag{1}$$

with kernel of reduction $\widehat{E}(p\mathbb{Z}/p^e)$ a p-group of order p^{e-1} [9]. The formal parameter $z = -X/Y$ identifies $\widehat{E}$ with $(p\mathbb{Z}/p^e, +_F)$ ($+_F$ the formal group law), and the formal logarithm $\log\colon \widehat{E} \to (p\mathbb{Z}/p^e, +)$ is an isomorphism with $\log([m]u) = m\log(u)$. For $E\colon y^2 = x^3 + b$ the invariant differential is $\omega = (1 + 3b\, z^6 + \cdots)\, dz$, so $\log(z) = z + \frac{3b}{7} z^7 + \cdots$; for $z \in p\mathbb{Z}$ and $e \leq 4$ the correction vanishes ($z^7 \in p^7\mathbb{Z}$), hence $\log(z) \equiv z \pmod{p^e}$ and kernel arithmetic is additive in z at our precision.

A *section* is a map $\tau\colon E(\mathbb{F}_p) \to E(\mathbb{Z}/p^e)$ with $\text{red} \circ \tau = \text{id}$; the Teichmüller and Hensel lifts are sections, generally not homomorphisms. We use the x-Teichmüller / Hensel section throughout; it satisfies $\tau(-P) = -\tau(P)$ (the two Hensel roots in y are negatives), a property we invoke once below. The *lift error* (a 1-cochain valued in $\widehat{E}$ by (1)) is

$$\delta_\tau(k) \;:=\; [k]\,\tau(G) \;-\; \tau(kG) \;\in\; \widehat{E}(p\mathbb{Z}/p^e). \tag{2}$$

We write $\delta(k)$ for the Teichmüller case and $z(\delta(k)) \in p\mathbb{Z}/p^e$ for its formal parameter; for an integer $z \in \mathbb{Z}/p^e$, $\text{digit}_j(z)$ is the coefficient of p^j in the base-p expansion (digit 0 is the units place, identically 0 here since $z \in p\mathbb{Z}$).

Prior art. That a p-adic lift transports the ECDLP without weakening it, that the canonical (Teichmüller) lift carries no information, and that "solving the DLP is equivalent to constructing a non-canonical lift" are due to Silverman [2, 1] for elliptic curves and to Gadiyar–Maini–Padma and Gadiyar–Padma [3] for $\mathbb{F}_p^\times$. Smart, Satoh–Araki and Semaev [4, 5, 6] give the anomalous attack; the canonical lift is Lubin–Serre–Tate [7], its computational instance Satoh's point counting [8]. Our theorem repackages these in the lift-error language.

3 The exploration, in brief

The full study ran forty-seven numbered phases against the canonical-lift / formal-group attack surface for $j = 0$ (CM by $\mathbb{Z}[\omega]$) and generic ordinary curves; Appendix A indexes them. Probes included pseudorandomness of δ (DFT flatness, maximal Berlekamp–Massey complexity, uniform p-adic valuation, no polynomial/Mahler/multiplicative fit); a homomorphic section from H^2 vanishing (which exists but, being a homomorphism, annihilates the attack); an explicit Serre–Tate canonical lift validated by $\delta_{\text{can}} \equiv 0$; HNP/lattice (LLL, BKZ) attacks on the cocycle; MOV embedding degree; ordinary/supersingular bridges; genus-2 Jacobian covers; Weil pairings; Iwasawa μ; and a Frobenius/unit-root analysis. Every one was inert. Two exact relations survived, both *public* (computable from the curve alone) and both giving no advantage: an automorphism-equivariance of the leading δ-digit, and the antisymmetry

$$\delta(k) + \delta(n-k) = [n]\,\tau(G), \tag{3}$$

which holds for any section with $\tau(-P) = -\tau(P)$ (so that $\tau(kG) + \tau(-kG) = O$), and which we confirmed holds *exactly* in the z-coordinate—e.g. for $p = 97$ ($n = 79$; a curve distinct from the $p = 67$, $n = 79$ entry of Table 1), $z(\delta(k)) + z(\delta(n-k))$ takes a single value across all 39 reflection pairs $(k, n-k)$.

4 A false positive: the Gowers anomaly

Encode digit j of the lift error as a phase, $f_j(k) = \exp\bigl(2\pi i\,\mathrm{digit}_j(z(\delta(k)))/p\bigr)$ for $k = 1, \ldots, n-1$, and measure the Gowers U^2 norm [10] against an S^1-uniform random baseline (definitions and counts in §6). Across many primes the U^2 norm was elevated 5–11σ on the original exploratory sweep (Phase 37, a mixed bag of curves later found to include supersingular and anomalous cases); on the clean controlled set of §5 the elevation is $\sim$3–14σ (z(real vs rand) ranges 3.3 to 13.6). U^3 and higher sat at baseline. The leading Fourier coefficient decayed as $|\hat{f}(\alpha)| \sim Cn^{-\beta}$ with $\beta < 1/2$, on CM, j=1728, and non-CM curves alike. An internal report concluded this was a genuine, previously unnoticed degree-1 structural property of the Teichmüller cocycle "not explained by any universal identity."

The evidence offered was a permutation control: randomly permuting the *values* of f_j destroyed the elevation, read as proof that the structure lived in the index ordering, hence was "real." This inference is invalid, and seeing why is the point of the paper.

5 Mechanism-derived controls

The principle. A control is a null model, and a null tests exactly one alternative: the one under which the statistic is recomputed. The value-permutation control recomputes U^2 under "the multiset of digit values, in random order," so it separates "structure in the value distribution" from "structure in the ordering"—and nothing else. But at least three mechanisms predict ordered structure here, and the permutation kills all three at once, so it cannot attribute the signal to any one:

(H1) *Prefix-sum artifact.* $\delta(k) = -\sum_{i<k} c(iG, G)$ is a running sum; summation is a low-pass filter ($\hat{s}(\xi) = \hat{d}(\xi)/(1 - e^{2\pi i\xi/n})$, a $1/\xi$ pole at $\xi = 0$), so *any* increments with this multiset, once summed, carry elevated low-frequency U^2.

(H2) *Antisymmetry shadow.* The exact reflection (3) couples Fourier mode α with $-\alpha$, a rigid constraint that elevates U^2.

(H3) *Genuine new structure*, the claimed explanation.

We build one null per mechanism, each derived from how that mechanism generates ordering. All numbers below are on the clean ordinary set of Table 1 (Phase 43), under the pre-registered protocol of §6.

H1 (increment shuffle). $s(k) = z(\delta(k))$ is the exact cumulative sum of its first differences $d(k) = s(k+1) - s(k)$. Shuffle d, re-sum in the new order, recompute U^2; the increment multiset is identical, only the order changes. The shuffled-increment prefix sums sit *at* the random baseline ($z(\text{incShuf vs rand}) \in [-0.3, +0.5]$ on every curve), while the true ordering sits +3.5 to +12.5 above them, growing with n (Table 1, cols. 4–5). H1 is refuted: the digit nonlinearity destroys the walk's low-frequency mass, so the signal is in the *specific* order, not in summation.

H2 (half-sequence). The reflection (3) relates each k to $n-k$. On the first half $k = 1, \ldots, \lfloor (n-1)/2 \rfloor$ every reflection partner lies outside the window, so H2 predicts no internal constraint there while H3 predicts the signal persists. Re-running the increment-shuffle control on the half collapses the elevation to noise ($z(\text{half vs incShuf}) \in [-1.3, +1.5]$, no growth in n; Table 1, col. 6).

H2, positive control (the decisive test). Absence of signal on the half is necessary but not sufficient. We therefore test H2 *constructively*: generate synthetic sequences that impose *only* the reflection, $s(k)$ i.i.d. uniform on the first half and $s(n-k) = C - s(k)$ with $C = z([n]\tau(G))$, carrying no other structure, and measure their U^2. If reflection alone is the mechanism, these must reproduce the real statistic. They do, quantitatively: the synthetic U^2 is elevated +3.8 to +12.5 over the random baseline—tracking the real elevation across the whole range of n—and the real sequences are statistically indistinguishable from them ($z(\text{real vs syn}) \in [-1.4, +1.5]$; Table 1, cols. 7–8). The match on the U^2 elevation and its growth with n is the substance of the positive control, and it is exact (cols. 7–8). The finer digit-dependent Fourier decay is only *partially* reproduced: fitting $|\hat{f}| \sim Cn^{-\beta}$ across the seven curves, the real sequences give $\beta_{\text{real}} \approx \{0.34, 0.45, 0.36\}$ for digits $1, 2, 3$, whereas the reflection-only synthetic draws give a nearly flat $\beta_{\text{syn}} \approx \{0.41, 0.41, 0.40\}$. Reflection alone thus reproduces the overall power-law decay ($\beta \approx 0.4$, the same order as the real exponents) but not its digit-to-digit variation; we do *not* claim the spectra match in detail. The residual digit-to-digit variation is, however, consistent with finite-size fit scatter rather than secondary structure: resampling the seven-curve fit on reflection-only sequences gives β_{syn} with mean ≈ 0.40 and standard deviation ≈ 0.06 *with no digit dependence built in*, and each real exponent $\{0.34, 0.45, 0.36\}$ lies within ≈ 1 standard deviation of that mean. The mild downward spread of the interior-digit exponents is moreover the *sign* predicted by the carry model of §5 (Prop. 2): digit 1 reflects exactly while digits $2, 3$ carry a small borrow-induced deficit. Under the standard assumption that the ECDLP is hard on these curves, the residual cannot be cryptanalytic in any case—it is a deterministic post-processing of the public digit map of a public object (Theorem 1, Corollary 1)—so we do not pursue it further; the decisive evidence that H2 is the mechanism is the U^2 match, which is genuine. A post-hoc confirmatory run on a large-prime-order curve ($p = 10477$, $n = 10639$ prime; Phase 47, last row of Table 1) extends the pattern by more than an order of magnitude in n: the real elevation reaches +43–44, the reflection-only synthetics track it at +44.5–44.8, and real vs. synthetic stays within ± 2.1.

The converse control (breaking the reflection). A positive control can be accused of circularity—a synthetic sequence built with the reflection will of course exhibit the reflection statistic. The decisive converse is to remove the reflection from the *real* construction and check that the elevation disappears. The antisymmetry (3) holds only because the x-Teichmüller section satisfies $\tau(-P) = -\tau(P)$. We therefore replaced it with an *asymmetric* section τ' that selects the Hensel y-root by a negation-insensitive rule (smaller integer representative), so $\tau'(-P) \neq -\tau'(P)$ in general; the identity $z(\delta'(k)) + z(\delta'(n-k))$ then ceases to be single-valued (verified: 0 of $\lfloor (n-1)/2 \rfloor$ pairs share a common sum, versus all of them for τ). On the same curves, same secrets, same digit encoding, the U^2 elevation collapses from z(real vs rand) of $+3.6, +6.5, \ldots, +45.9$ to baseline ($+1.4, +0.4, \ldots, -1.8$ across $p = 67, 211, 823, 10477$; Phase 49). Breaking $\tau(-P) = -\tau(P)$ destroys both the identity and the elevation, which is exactly the prediction if the reflection—and nothing else about δ—generates the anomaly. (With τ' exactly half the lift errors leave the kernel of reduction, so the pre-registered filter halves the usable length; the length-independent kill signal is the loss of single-valuedness, and we report the length-matched comparison alongside in the repository.)

The growth law is itself an artifact signature. The elevation does not merely persist with n; it *grows*, from ~ 3 at $n = 79$ to ~ 44 at $n = 10639$. It is tempting to read growth-with-n as accumulating signal—this is precisely what the internal Phase-37 report did. The opposite is true: the growth is forced by the reflection alone and carries no curve information, as the following closed form shows.

We treat the general mechanism first and read off the reflection as the extremal case. Write the U^2 mass as $\|f\|_{\mathrm{U}^2}^4 = \sum_\xi |\hat{f}(\xi)|^4$ with $\hat{f}(\xi) = \frac{1}{L}\sum_k f(k)e^{-2\pi i k\xi/L}$. In the large-$L$ limit each nonzero mode $Z_\xi := \hat{f}(\xi)$ is a mean-zero complex Gaussian characterized by two second moments: the *variance* $\sigma^2 = \mathbb{E}|Z_\xi|^2 = 1/L$ and the *pseudo-variance* $\rho(\xi) = \mathbb{E}[Z_\xi^2]$, the latter zero for a proper (circularly symmetric) mode and nonzero exactly when a constraint couples ξ to $-\xi$.

Proposition 1 (Pseudo-variance classification of U^2 inflation)**.** *For a complex Gaussian Z with variance σ^2 and pseudo-variance ρ,*

$$\mathbb{E}|Z|^4 = 2\sigma^4 + |\rho|^2. \tag{4}$$

Hence for $f\colon \mathbb{Z}/L \to S^1$ whose modes have per-mode pseudo-variance $\rho(\xi)$, $\mathbb{E}\big[\|f\|_{\mathrm{U}^2}^4\big] = \big(2 + \langle|\rho|^2\rangle/\sigma^4\big)/L + o(1/L)$, where $\langle\cdot\rangle$ averages over the $L-1$ nonzero modes, and the mass-inflation factor relative to the i.i.d. baseline is

$$\mathit{INFLATION} \;=\; 1 + \frac{\langle|\rho(\xi)|^2\rangle}{2\sigma^4} \;\in\; [1, \tfrac{3}{2}]. \tag{5}$$

The upper bound $|\rho| \le \sigma^2$ (Cauchy–Schwarz) is attained, giving the maximal $3/2$, iff *$|\rho(\xi)| = \sigma^2$ a.e.—the case of an index-reversing constraint $\hat{f}(-\xi) = \beta\,\overline{\hat{f}(\xi)}$; a constraint that fixes each mode (e.g. a translation $k \mapsto k+s$) gives $\rho \equiv 0$ and no inflation.*

Proof. Write $Z = X + iY$ with real covariance $\left(\begin{smallmatrix} a & c \\ c & b \end{smallmatrix}\right)$, so $\sigma^2 = a + b$ and $\rho = (a-b) + 2ic$, $|\rho|^2 = (a-b)^2 + 4c^2$. Isserlis' theorem gives $\mathbb{E}[X^4] = 3a^2$, $\mathbb{E}[Y^4] = 3b^2$, $\mathbb{E}[X^2Y^2] = ab + 2c^2$, whence $\mathbb{E}|Z|^4 = \mathbb{E}[(X^2+Y^2)^2] = 3a^2 + 3b^2 + 2ab + 4c^2 = 2(a+b)^2 + (a-b)^2 + 4c^2 = 2\sigma^4 + |\rho|^2$, proving (4). Summing over modes and using $\sigma^2 = 1/L$ gives the mass and (5); the bound $|\rho| \le \sigma^2$ is Cauchy–Schwarz on $\mathbb{E}[Z^2]$ versus $\mathbb{E}|Z|^2$. The index-reversing constraint forces $Z_{-\xi} = \beta\,\overline{Z_\xi}$, hence $\mathbb{E}[Z_\xi^2]$ has modulus $\mathbb{E}|Z_\xi|^2 = \sigma^2$; a constraint fixing each mode leaves $\mathbb{E}[Z_\xi^2] = 0$. $\square$

Proposition 2 (The reflection inflates the U^2 mass by exactly $3/2$)**.** *Let $f\colon \mathbb{Z}/L \to S^1$ be subject to* only *the antisymmetry constraint $f(L{+}1{-}k) = \beta\,\overline{f(k)}$ ($\beta \in S^1$ fixed, the free half i.i.d. S^1-uniform)—the reflection-only synthetic of the positive control. Then $\mathbb{E}\big[\|f\|_{\mathrm{U}^2}^4\big] = 3/L + o(1/L)$, versus $2/L + o(1/L)$ for i.i.d. S^1-uniform entries; the ratio is exactly $3/2$, independent of L and of the value alphabet. The standardized synthetic-vs-random gap is $z(\mathrm{syn/rand}) = 2^{7/4}\big(3^{1/4} - 2^{1/4}\big)\sqrt{L} + o(\sqrt{L}) \approx 0.427\sqrt{L}$.*

Proof. The constraint is index-reversing, so by Prop. 1 the inflation factor is $3/2$, i.e. $\mathbb{E}[\|f\|^4_{\mathrm{U}^2}] = 3/L$. For the standardized gap, the i.i.d. mass has standard deviation $\Theta(L^{-3/4})$; with the normalization above its leading constant is $\mathrm{SD} = 2^{-7/4}L^{-3/4}$, while the mean gap in the U^2 *norm* (the fourth root) is $(3/L)^{1/4} - (2/L)^{1/4} = (3^{1/4} - 2^{1/4})L^{-1/4}$. Their ratio is $2^{7/4}(3^{1/4} - 2^{1/4})\sqrt{L}$. □

The coefficient $2^{7/4}(3^{1/4} - 2^{1/4}) = 0.4267\ldots$ is fully analytic; it is the closed form of the constant the original sweep had only measured. Standardizing the U^2 *norm* against the empirical-null standard deviation, we confirmed it p- and L-independent across $L \in [78, 10^6]$ (Phases 48, 56): the mass ratio holds to 1.500 at $L = 10^6$, and $0.427\sqrt{n}$ reproduces the syn/rand column of Table 1 and the Phase-47 value (predicted 44.8 vs. observed 44.6 at $n = 10639$) within finite-size noise. So $\sqrt{n}$ growth is the *signature of the antisymmetry artifact*, not of an accumulating signal—the exact inference the original report inverted. Equation (5) also makes the discriminator quantitative: a constraint sending mode ξ to its reversal $-\xi$ inflates the mass to $3/L$, whereas one sending it elsewhere (e.g. the fixed-shift involution $k \mapsto k + L/2$) leaves it at $2/L$; we verified both numerically (ratios 1.499 vs. 1.001; Phase 55), so an auditor reads the mechanism off the constraint's action on Fourier indices.

Higher norms, and a carry prediction. Two consequences of the same analysis sharpen the picture. First, the inflation is a purely second-order (quadratic-mass) phenomenon: the extra difference operator defining U^3 couples reflection partners only on an $O(1/L)$ fraction of shift–position pairs, so $\mathbb{E}\|f\|^8_{\mathrm{U}^3}$ sits at the unconstrained baseline up to $O(1/L)$. This *derives* the otherwise unexplained observation that U^3 and higher sat at baseline; we confirmed the U^3 excess decays as $1/L$ while the U^2 ratio holds at $3/2$ (Phase 59). Second, the real digit sequence reflects only approximately: base-p borrows mean the phase-level reflection $\mathrm{digit}_j(C-s) \leftrightarrow \mathrm{digit}_j(s)$ holds for a fraction $1 - q_j$ of pairs. Modelling the rest as independent gives per-mode $|\rho| = (1 - q_j)\sigma^2$, hence by (5) a mass $\big(2 + (1 - q_j)^2\big)/L$. Digit 1 reflects *exactly* ($q_1 = 0$, since $z(\delta) \in p\mathbb{Z}$ forces digit 0 to vanish on both sides, so no borrow reaches position 1); interior digits have $q_2, q_3 \approx 0.2$ measured on the clean set, predicting a mild mass deficit there. At the clean-set sizes this deficit (~ 0.3 in L·mass) sits at the $\lesssim 2\sigma$ resolution limit and is *not* yet distinguishable from finite-size scatter—consistent with the small β_{real} spread we report below—so we state it as a falsifiable prediction: a confirmatory run at $L \gtrsim 3000$ should resolve U^2 masses of ≈ 3.0 at digit 1 and ≈ 2.7 at digits $2, 3$ (Phase 58).

Verdict. The anomaly is the Fourier-domain shadow of the exact, public identity (3): H2, not H1 or H3. It is real but not new, and—being a public consequence of the curve's geometry that yields only a 2-to-1 search reduction—carries no cryptanalytic advantage. The varying "decay exponent" and its apparent curve-family dependence in the internal report are finite-size scatter around a single reflection-induced spectrum; nothing family-specific survives. To rule out a complex-multiplication artifact—the clean set is $j{=}0$, which carries extra automorphisms—we re-ran the full protocol on $j{=}1728$ and on generic non-CM curves ($j \in \{6, 316, 446\}$, prime order, $\gcd(n, p) = 1$): the 3/2 inflation, the syn/rand tracking, and the real-vs-syn equivalence reproduce identically, confirming the mechanism is the antisymmetry and not the CM structure (Phase 50).

6 Statistical protocol

All quantities in §5 follow one pre-registered protocol (fixed before the clean re-run; seed 20260609). Precision $e = 4$; digits $j \in \{1, 2, 3\}$. The S^1-random baseline draws 300 sequences $g(k) = e^{i\theta_k}$, θ_k uniform on $[0, 2\pi)$, of the same length L; "σ" and "z" are the standard deviation and standardized

p	n	L	negative controls (z) real/inc	inc/rand	half/inc	positive (z) syn/rand	real/syn
67	79	78	$+3.5..6.7$	$-0.3..0.2$	$-0.6..1.5$	$+3.8..4.1$	$-0.4..1.2$
163	139	138	$+3.7..7.4$	$+0.0..0.5$	$-0.4..1.0$	$+5.4..5.4$	$-0.9..1.5$
211	199	198	$+5.1..7.8$	$+0.1..0.3$	$-0.8..0.5$	$+5.8..5.9$	$-0.3..0.8$
349	313	312	$+5.4..7.1$	$-0.0..0.1$	$-1.3..0.2$	$+7.5..7.6$	$-1.4..-0.4$
433	397	396	$+8.2..9.6$	$-0.1..0.2$	$-0.4..0.6$	$+8.4..8.6$	$-0.1..0.7$
577	613	612	$+7.8..11.5$	$+0.1..0.2$	$-0.5..1.1$	$+10.2..10.4$	$-1.2..0.8$
823	829	828	$+10.5..12.5$	$+0.1..0.2$	$-0.9..0.0$	$+12.2..12.5$	$-0.9..0.7$
$10477^{\dagger}$	10639	10638	$+42.7..44.1$	$-0.0..0.0$	$-1.1..1.8$	$+44.5..44.8$	$-2.0..0.7$

Table 1: Phase 43, clean ordinary j=0 curves $y^2 = x^3 + 7$ ($p \equiv 1 \bmod 3$, $n = \#E(\mathbb{F}_p)$ prime, so $\gcd(n, p) = 1$ and the curve is neither supersingular nor anomalous); L is the number of usable $k \in [1, n-1]$ after the pre-registered degeneracy filter (§6); here $L = n - 1$ (no drops). Ranges are over digits $j = 1, 2, 3$. Negative controls: shuffling increments leaves U^2 at the random baseline (inc/rand) and the true ordering far above it (real/inc), which vanishes on the half (half/inc). Positive control: synthetic reflection-only sequences reproduce the elevation (syn/rand) and the real sequences match them (real/syn). The entire effect is the antisymmetry (3). The last row (†, Phase 47, $p = 10477$) is a post-hoc confirmatory run under the identical protocol, reported separately from the pre-registered seven-curve set.

deviation *of the relevant null*: incShuf uses 300 increment shuffles, syn uses 300 synthetic draws, rand uses the 300 baselines. The Gowers U^2 is computed exactly by its FFT formula. *Exclusion rule (pre-specified):* a value $z(\delta(k))$ is dropped iff its projective lift is degenerate—Y a non-unit, or $z \not\equiv 0 \bmod p$ (a kernel element must have $z \equiv 0$); these are arithmetic failures of the pure-Python lift, flagged automatically, and on the clean ordinary set none occurred ($L = n - 1$ throughout). The clean set is the seven ordinary j=0 curves $y^2 = x^3 + 7$ with $p \equiv 1 \bmod 3$ and $n = \#E(\mathbb{F}_p)$ prime (hence $\gcd(n, p) = 1$, neither supersingular nor anomalous); the smallest has $n = 79$, so $L \geq 78$ throughout. No further size-based filter is applied—every such curve in the run is reported. We make no multiplicity correction and therefore do *not* claim discovery from any single elevated U^2; the paper's logic is the opposite—every elevation is traced to a named mechanism and shown to be identity-or-artifact, and the controls are designed to *remove* signal, not to find it. Phase 47 repeats this protocol verbatim on one large-prime-order curve ($p = 10477$, $n = \#E(\mathbb{F}_p) = 10639$ prime); being post-hoc, it is reported separately (marked † in Table 1). Throughout, z-values are standardized effect sizes against 300-draw empirical nulls, not calibrated Gaussian tail probabilities; we use them only comparatively, never as discovery statistics.

Calibration. Three checks make the evidential language precise. (i) *Look-elsewhere.* The exploration spans $\sim$47 phases $\times$ 3 digits $\times$ 8 curves $\times$ several encodings; the expected maximum $|z|$ of a generic random baseline over a battery of this size is ≈ 3.2–3.6 by extreme-value theory, and an empirical 10-statistic battery on i.i.d. surrogates reached $|z| = 5.0$ (Phase 54). A "5σ" separation from a generic baseline is therefore *expected* under this much exploration—which is the paper's point; the syn/rand elevations of 10–44 sit far above this floor only because they are a real (public, non-cryptanalytic) effect. (ii) *Equivalence, not absence of significance.* For the real-vs-synthetic comparison we report a two-one-sided-test bound: with 300 synthetic draws the data exclude any residual non-reflection ("H3") component larger than a curve-dependent fraction of the elevation, tightening from ≈ 0.9 at the smallest curve to ≈ 0.25 at $p = 823$ (Phase 52). Equivalence is thus established tightly only at larger L; we say "indistinguishable" in that bounded sense. (iii) *Power of the half-sequence control.* Because the gap grows as $\sqrt{L}$, an intrinsic (length-extensive) H3 signal of the observed full-length size would still appear at $\approx z_{\text{full}}/\sqrt{2}$ on the half. At the half length we retained 4–9σ of power to detect it and observed ≈ 0, excluding an intrinsic H3 on 6 of the 7 curves; only the smallest ($p = 67$) is genuinely underpowered (Phase 53).

7 Why it had to be a false positive: an inertness theorem

For $\gcd(n, p) = 1$ the secret-dependent part of any section lift error is forced into a coordinate the attacker cannot compute, so any "structure" in δ that *is* computable from (E, G, Q) must reduce to a public identity or an artifact. We make this precise.

Lemma 1 (Canonical lift)**.** *If* $\gcd(n, p) = 1$*, the order-*n *subgroup* $\langle G\rangle$ *lifts uniquely to* $\langle \widetilde{G}\rangle \subset E(\mathbb{Z}/p^e)$ *of order* n *with* $\mathrm{red}(\widetilde{G}) = G$*, and* $jG \mapsto [j]\widetilde{G}$ *is a homomorphism (so* $\delta_{\tau_{\text{can}}} \equiv 0$*). Explicitly* $\widetilde{G} = [m]\tau(G)$ *for any section* τ*, where* $m \equiv 1 \pmod n$*,* $m \equiv 0 \pmod{p^{e-1}}$*.*

Proof. $E(\mathbb{Z}/p^e)[n]$ injects into $E(\mathbb{F}_p)[n]$ (an n-torsion element of $\widehat{E}$ is killed by $\gcd(n, p^{e-1}) = 1$, hence trivial) and surjects onto it (étale n-torsion lifts by Hensel), so reduction is an isomorphism on n-torsion and $\widetilde{G}$ is the unique order-n lift of G. Writing $\tau(G) = \widetilde{G} + c$ with $c \in \widehat{E}$, $[m]\tau(G) = [m \bmod n]\widetilde{G} + [m \bmod p^{e-1}]c = \widetilde{G}$. □

Theorem 1 (Inertness)**.** *Let* τ *be any section and* $c := \tau(G) - \widetilde{G} \in \widehat{E}$*. Then:*

(a) $\delta_\tau(k) = [k]c - c_k$*, where* $c_k := \tau(kG) - [k]\widetilde{G} \in \widehat{E}$ *depends only on the point* kG*.*

(b) $[k]c$ *depends on* k *only through* $k \bmod p^{e-1}$*;* $\log([k]c) = k \log(c)$ *has additive order a power of* p *dividing* p^{e-1}*.*

(c) $\gcd(n, p^{e-1}) = 1$*, so by CRT, for* k *uniform on* $\mathbb{Z}/np^{e-1}\mathbb{Z}$ *the residues* $k \bmod n$ *(fixed by* Q*) and* $k \bmod p^{e-1}$ *(governing* $[k]c$*) are independent. The public data* (E, G, Q) *fixes only* $k \bmod n$*; hence over this natural lift* $[k]c$*, a function of* $k \bmod p^{e-1}$*, is not determined by the public data.*

(d) δ_τ *is a well-defined function of the instance (i.e. of* $k \bmod n$*) iff* $c = 0$ *(i.e.* τ *canonical on* $\langle G\rangle$*), in which case* $\delta_\tau \equiv 0$*. For* $c \neq 0$*,* $\delta_\tau(k+n) - \delta_\tau(k) = [n]c = [n]\tau(G) \neq O$*, the antisymmetry constant of* (3)*.*

Proof. (a) $[k](\widetilde{G} + c) - ([k]\widetilde{G} + c_k) = [k]c - c_k$. (b) $c \in \widehat{E}$ has order dividing p^{e-1}; apply log, whose order as a map $k \mapsto k \log(c)$ on $(p\mathbb{Z}/p^e, +)$ is $p^e/\gcd(\log(c), p^e)$, a power of p dividing p^{e-1} since $\log(c) \in p\mathbb{Z}$. (c) Since $\gcd(n, p^{e-1}) = 1$, CRT gives $\mathbb{Z}/np^{e-1}\mathbb{Z} \cong \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/p^{e-1}\mathbb{Z}$, so for k uniform on the left the residues $k \bmod n$ and $k \bmod p^{e-1}$ are independent; as (E, G, Q) determines only $k \bmod n$, it leaves $k \bmod p^{e-1}$ (and hence $[k]c$) information-theoretically unconstrained over this lift. We make no claim of a reduction beyond this. (d) c_k is a function of kG, hence of $k \bmod n$; so δ_τ factors through $k \bmod n$ iff $k \mapsto [k]c$ does, iff $[n]c = O$, iff (as $\gcd(n, p^{e-1}) = 1$) $c = O$. The shift equals $[n]c = [n]\tau(G)$ since $[n]\widetilde{G} = O$. □

Corollary 1. *For* $\gcd(n, p) = 1$ *no section lift error yields an efficiently computable function of* (E, G, Q) *that depends on the secret: it is either identically zero (canonical) or not a function of the instance at all (non-canonical, ambiguity* $[n]\tau(G) \neq O$*). The* p*-adic canonical lift moves the ECDLP without weakening it.*

Generality. The proof uses nothing about elliptic curves beyond the exact sequence (1): an extension $0 \to K \to B \to G \to 0$ of abelian groups with G finite of order n, K a p-group of exponent dividing p^{e-1}, and $\gcd(n, p) = 1$. Then n is invertible on K, so $H^2(G, K) = 0$, the retraction $b \mapsto \mu([n]b)$ with $\mu = ([n]|_K)^{-1}$ splits the sequence, and the lift error is a coboundary carrying no extension data. Theorem 1 is the instance $B = E(\mathbb{Z}/p^e)$, $G = E(\mathbb{F}_p)$, $K = \widehat{E}(p\mathbb{Z}/p^e)$; because the hypotheses are purely group-theoretic it ports verbatim to Jacobians of higher-genus curves, algebraic tori, and any commutative group scheme with smooth reduction and fibre order coprime to p. This is what we mean by "regime-agnostic": the conclusion is a one-line consequence of coprime-order cohomology vanishing, independent of the curve sizes tested.

This is why the Gowers anomaly was forced to be artifact-or-identity. The only secret-bearing object, $[k]c$, is not publicly computable; the visible δ, taken as a function of the integer k, is either zero or periodic-with-shift $[n]\tau(G)$—and that shift is the reflection (3) whose Fourier signature the probe rediscovered.

Remark (on clause (c)). We deliberately do *not* claim $[k]c$ is information-theoretically independent of (E, G, Q): once a canonical representative $k \in [1, n-1]$ is fixed, $k \bmod p^{e-1}$ and hence $[k]c$ are determined. The claim is computational—no efficient algorithm produces $[k]c$ from the public data short of solving the ECDLP—which is what matters for security and what the experiments probe. Being a state-of-knowledge assertion, this claim is confined to the present remark rather than the theorem statement.

8 The boundary is sharp

Theorem 1 hinges on $\gcd(n, p) = 1$, and what changes at the boundary is the action of $[n]$ on the kernel of reduction. Set $n = p$ (an anomalous curve, $\#E(\mathbb{F}_p) = p$; take $e = 2$): $[n] = [p]$ now *annihilates* the kernel—$[p]c = O$ for every $c \in \widehat{E}(p\mathbb{Z}/p^2)$, since $\log([p]c) = p\log(c) \in p^2\mathbb{Z}$—whereas for $\gcd(n, p) = 1$ it acts as an automorphism. Two consequences follow. First, the *section ambiguity* dies: two sections differ pointwise by a kernel element, which $[p]$ kills, so $[p]\tau(P)$ is independent of the choice of section. Second, what survives is *not* zero: $\varphi(P) := \frac{1}{p}\log\big([p]\tau(P)\big) \bmod p$ is a homomorphism $E(\mathbb{F}_p) \to \mathbb{F}_p$, linear in the secret, and Smart's attack is its evaluation, $k = \varphi(Q)/\varphi(G)$. The asymmetry with the ordinary case is exact: there the ambiguity has maximal order and the only well-defined quantity is the public constant of (3); here the ambiguity vanishes and the surviving invariant carries k. (Note that δ itself remains ill-defined as a function of $k \bmod p$: $\delta(k+p) - \delta(k) = [p]\tau(G) \neq O$. The object that becomes well-defined is $[p]\tau(P)$, not δ.) We verified both sides (Table 2): on every ordinary test curve the obstruction $[n]\tau(G)$ is a nonzero kernel element of *maximal* order p^{e-1} (valuation 1), while on the anomalous $y^2 = x^3 + 4x + 7$ over $\mathbb{F}_{53}$ (with $G = (0, 22)$) the value $z([p]\tau(G)) = 15 \cdot 53 \bmod 53^2$ is *section-independent* (identical under the Teichmüller-x, naive-x, and shifted-x sections) and nonzero—as it must be, since the attack divides by $\varphi(G) = 15$—and Smart's attack recovers k for *all* 52 nonzero residues $k \in [1, p-1]$ (52/52; an exhaustive implementation validation, Phase 51). The same code path that is inert for $\gcd(n, p) = 1$ is a total break the instant $n = p$.

The dichotomy is exhaustive, not merely sharp. One might ask whether an *intermediate* regime exists—a curve on which $[n]$ partially degenerates on the kernel, $0 \neq \ker([n]|_{\widehat{E}}) \subsetneq \widehat{E}$, giving a graded leak. On $E(\mathbb{F}_p)$ there is none. A partial kernel needs $1 \leq v_p(n) < e-1$ with $v_p(n) \geq 1$, i.e. $p \mid n$; but $n \mid N = \#E(\mathbb{F}_p)$, so $p \mid n$ forces $p \mid N$, and by Hasse $N \in [(\sqrt{p}-1)^2, (\sqrt{p}+1)^2]$ the only multiple of p in range is $N = p$ itself for $p \geq 7$ (we verified no curve with $p \mid N \neq p$ exists

Regime	curves	$[n]$ on $\widehat{E}$	invariant $[n]\tau(G)$	ECDLP
Small ordinary, $\gcd(n,p)=1$	$p=31$–163, $j{=}0$	automorphism	$\neq O$, ord $= p^{e-1}$, public	inert
Anomalous (Smart), $n=p$	$y^2{=}x^3{+}4x{+}7/\mathbb{F}_{53}$	zero map	$\neq O$, section-indep., $\propto k$	broken, 52/52

Table 2: Phases 41–42. What vanishes at the boundary is the *ambiguity*, not the invariant. For $\gcd(n,p)=1$, $[n]$ is an automorphism of the kernel: the well-definedness obstruction of Theorem 1(d) has maximal order and the only well-defined quantity is the public antisymmetry constant of (3). For $n=p$, $[n]$ annihilates the kernel: the section ambiguity $[p]c$ dies and $[p]\tau(G)$ becomes a well-defined, *nonzero*, secret-linear invariant—which Smart's attack requires ($\varphi(G)\neq 0$). $\gcd(n,p)=1$ is the exact boundary. Theorem 1 is regime-agnostic—it holds for any $\gcd(n,p)=1$—so these small ordinary curves (chosen for exact-arithmetic reach) suffice to verify it; the relevance to large-prime ECDSA curves is through the theorem's generality, not through these particular orders.

for $p=7,\dots,23$; the lone exception $N=2p$ occurs only at $p=5$). Hence $v_p(n)\in\{0,1\}$ and $[n]$ acts on $\widehat{E}$ as an automorphism ($p\nmid n$) or, on its p-part, the zero map ($n=p$), never partially. The boundary is a true dichotomy with no curves in between; a "graded inertness theorem" has empty interior here.

9 Numerical verification

Theorem 1 was checked on $y^2=x^3+7$ at $e=4$ for ten small ordinary curves over $\mathbb{F}_p$, $p=31,\dots,163$, all with $\gcd(n,p)=1$ (Phase 41). These test orders are deliberately small and several are composite—$n\in\{21,31,79,67,79,111,129,133,86,139\}$, five of them composite—chosen for exact-arithmetic reach rather than to model the large-prime ECDSA regime; since the theorem is regime-agnostic (it needs only $\gcd(n,p)=1$), they suffice to verify it. (Here n denotes the order of the chosen G and may be a proper divisor of $N=\#E(\mathbb{F}_p)$, e.g. $n=86$ with $N=172$ at $p=157$; the clean set of Table 1 instead requires $n=N$ prime.) Phase 47 additionally verifies every clause on one large-prime-order ordinary curve ($p=10477$, $n=N=10639$ prime): C_n again has valuation 1 and additive order exactly p^{e-1}, the antisymmetry is single-valued across all 5319 reflection pairs, and $L=n-1$ with no degenerate drops. All kernel arithmetic is done through the additive z-coordinate to avoid projective degeneracy on deep kernel points. On every prime: the obstruction $C_n := z([n]\tau(G))$ is nonzero in $p\mathbb{Z}$ of valuation 1 and additive order exactly p^{e-1} (clauses (d),(b)); the antisymmetry $z(\delta(k))+z(\delta(n-k))\equiv C_n$ holds for all k (clause (d) / (3)); the integer-representative shift $z(\delta(k+n))-z(\delta(k))\equiv C_n$ holds on all eight arithmetically-reliable primes (a group identity; on two primes the deep multiply degenerates in exact arithmetic, flagged by a self-consistency probe); and $\gcd(n,p^{e-1})=1$ (clause (c)). The canonical case $\delta_{\text{can}}\equiv 0$ (Lemma 1) was confirmed in an exact p-adic computation. Anomalous recovery is Phase 42 (§8).

10 Discussion

For negative cryptanalysis. The reusable lesson is narrow and practical. (i) A control tests exactly one alternative—the one its null recomputes; before trusting it, enumerate the mechanisms that could generate the statistic and check whether the control separates them. The permutation test here failed because three mechanisms collapsed under it. (ii) Build one null per mechanism, derived from how that mechanism generates the statistic: H1 from the prefix-sum structure (shuffle increments, re-sum), H2 from the reflection's reach (restrict to a half), and—decisively—a *positive* control synthesizing the surviving mechanism (reflection-only sequences) and showing it reproduces the statistic. (iii) Beware cumulative statistics: any running sum manufactures low-frequency autocorrelation, so "looks autocorrelated" is the null, not the finding. (iv) Where a structural theorem is available it bounds what controls can find; Theorem 1 says no honest control turns δ into an oracle for $\gcd(n, p) = 1$.

For automated and AI-assisted exploration. Pipelines that fan out many probes and surface the most anomalous are false-positive engines: with enough encodings and transforms an apparent 5–11σ effect is cheap, and a plausible-sounding control can launder it into a "finding." The safeguard is not more probes but mechanism-derived nulls, a positive control for any surviving signal, and—where possible—a closed-form reason the probe must be inert. The theorem's value here is exactly that: it turns "we tried forty things and they failed" into "here is the one-line reason they had to."

Honest scope. Theorem 1 is not new mathematics. It is an explicit instance, in the lift-error language, of the principle of [2, 1, 3] that lifting transports the ECDLP; the anomalous degeneration is Smart's attack [4]. We claim novelty only for the methodology—the mechanism-derived controls, the positive and converse controls that localize the anomaly, and the closed-form pseudo-variance account of the artifact (Prop. 1)—and for the explicit identification of the obstruction with the antisymmetry constant. The precise boundary of what this refutes is drawn next.

11 What this refutes, what it does not, and how to search next

The killed schema. The result terminates one specific attack schema: secret extraction from deterministic statistics of section lift errors $\delta_\tau(k)$—digit, phase, Fourier/Gowers, prefix-sum, spectral, recurrence, or machine-discovered structure in the ordered sequence $(k, \delta_\tau(k))$—on ordinary curves with $\gcd(n, p) = 1$. By Theorem 1(c,d) and Corollary 1, any such signal computable from (E, G, Q) carries a structural presumption of identity-or-artifact. We state the operative claim in the defensible form: this class of statistics has no *well-defined* secret-dependent content available from public data except at the anomalous boundary; under the standard assumption that the ECDLP is hard on ordinary non-anomalous curves, observed structure in these statistics should be presumed a public identity or an artifact unless it survives section-invariance and mechanism-derived controls. Appendix A is the empirical census of this schema; the theorem is the reason the census came back empty. Table 3 summarizes.

What this does not refute. The claim is bounded. It establishes *nothing* about: nonce-leakage and side-channel attacks on ECDSA; invalid-curve and twist attacks; small-subgroup attacks; small-embedding-degree / pairing (MOV) mechanisms, which are a separate functional inert here only for the independent reason of huge embedding degree; quantum algorithms; deliberately weakened curves; extension-field cases $E/\mathbb{F}_{p^m}$, where the Hasse dichotomy of §8 can fail; and—most relevant for the lifting literature—non-section-based global lift constructions. We claim no impossibility theorem for all p-adic or global lifts; in Silverman's "four faces" taxonomy [2, 1] the killed schema is one cell, not the map, and the anomalous boundary is precisely where a genuine p-adic attack (Smart) lives—we recover it, we do not contradict it.

A protocol for future searches. A credible new direction in this area should, before experimentation: (i) name the public invariant it computes from (E, G, Q); (ii) show it is well-defined

Killed schema	Why inert	Boundary / not covered
Statistics of $\delta_\tau(k)$ and its digit/phase/Fourier encodings, ordinary $\gcd(n, p) = 1$	Thm. 1: secret part $[k]c$ not publicly determined; visible δ is identity-or-artifact ((3))	$n = p$: Smart's attack (§8). Not covered: side channels, pairings, twists, quantum, $\mathbb{F}_{p^m}$, non-section global lifts

Table 3: Scope of the refutation. The schema is structurally dead on ordinary curves; the reusable contribution is the mechanism-resolved falsification protocol, not a broad impossibility claim.

independent of the section choice (the Theorem 1(d) test); (iii) show it is not forced to be canonically zero (Lemma 1) or a public identity (the fate of (3)); (iv) pair every candidate signal with mechanism-derived nulls—one for summation/walk artifacts, one for algebraic symmetries (reflection foremost), one for automorphism and coordinate-choice effects—*plus* a positive control synthesizing the surviving mechanism and, where possible, the converse test of breaking it; and (v) treat as evidence only a signal that survives all of these, scales on large ordinary prime-order curves, and admits a structural explanation not reducible to known lift inertness. The growth law (5) is a ready discriminator: $\sqrt{n}$ growth toward the 3/2 ceiling is the fingerprint of an index-reversal identity, not of an accumulating leak.

12 Reproducibility

Because the paper's thesis is that controls can mislead, its own workflow is held to a higher standard. All numbers are regenerated by the `ecdsa-lift` repository at the commit tagged with this paper (`674750a`); Python 3.10, NumPy 2.2, SymPy 1.14. Each phase is a standalone script under `experiments/` writing a JSON record under `results/`; `run_all.sh` reproduces Table 1 and the Proposition figures from the fixed seed 20260609. The pre-registered protocol (§6) was fixed before the clean re-run. Results fixed at submission are Phases 1–43 and 47–48; the converse, non-CM, all-k, calibration, and closed-form phases added in revision are 49–59, each cross-referenced above.

13 Conclusion

A long negative exploration produced one seductive anomaly. A standard control endorsed it; mechanism-derived controls, capped by a positive control, killed it and attributed it entirely to an exact public identity; and an inertness theorem explains why no such anomaly on ordinary curves can be more than artifact-or-identity, with the boundary falling exactly at the anomalous curves where the lift attack genuinely works. The cryptanalytic verdict is the expected negative one. The transferable result is a discipline: name your alternative hypotheses, build a null for each from its mechanism, synthesize the survivor to confirm it, and prefer a structural reason for inertness over an accumulation of passed tests.

A Phase index

Phase(s)	Target	Verdict
1–2	injectivity / pseudorandomness of δ	injective, pseudorandom
3–4	Smart–SSSA; homomorphic section from H^2	section exists, attack does not
5–6	CM-linearization; HNP on cocycle	pseudorandom; no bias
7–10	MOV; poly fit; LLL coords; ord/ss bridge	inapplicable / no leak
11–13	genus-2 Jacobian; GLV-CRT; Weil pairing	no new lever
14	$\mathrm{Aut}(E)$-equivariance rank law of δ-digit	public, no advantage
15–20	triple cocycle; DFT; Berlekamp–Massey; valuation; mult.	all inert
21b	antisymmetry $\delta(k) + \delta(n-k) = [n]\tau(G)$	exact, public
22–29	coord change; formal log; Mazur–Tate; Mahler	no structure
30–36	Sage validation; 2-cocycle; canonical lift; Iwasawa	inert
37	Gowers U^2 anomaly	false positive (this paper)
40,40b	increment-shuffle and half-sequence controls	localizes to 21b
41	numerical verification of Theorem 1	confirmed
42	anomalous boundary (Smart)	52/52 recovery
43	clean-ordinary re-run + positive control	confirms §5
44–46	aggregation/MI estimators; alt. uniformizers; OEIS scan	post-submission, see repository
47	large-prime-order confirmatory curve ($p = 10477$)	confirms §5, §9
48	U^2 growth law (Prop. 2: 3/2 mass, $\sqrt{n}$)	artifact signature
49	asymmetric-section converse control	elevation collapses
50	non-CM / j=1728 curves	pattern CM-independent
51	Smart attack, all $k \in [1, p-1]$	52/52 recovery
52–53	TOST equivalence; half-sequence power	residual/power bounded
54	look-elsewhere battery on surrogates	5σ is cheap
55–56	fixed-shift discriminator; large-L (10^6) constant	1.001 vs 1.499; 0.427
57	pseudo-variance classification (Prop. 1)	$\mathbb{E}\|Z\|^4 = 2\sigma^4 + \|\rho\|^2$
58	carry model q_j; resolvability $\sim \sqrt{L}$	prediction at $L \gtrsim 3000$
59	U^3 inflation $\to 1$ (excess $\sim 1/L$)	confirms baseline

Table 4: Index of the exploration. Full scripts, fixed seeds, and JSON outputs are in the `ecdsa-lift` repository.

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